

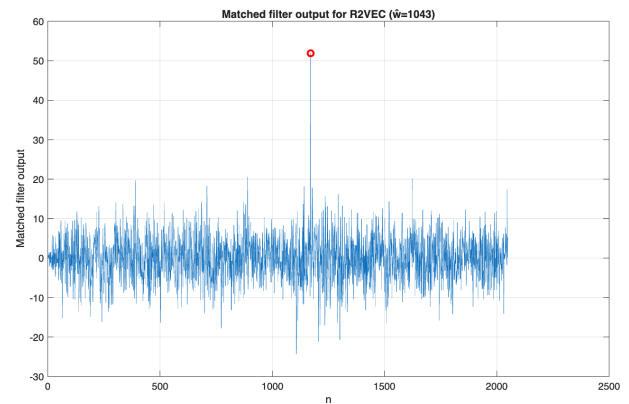
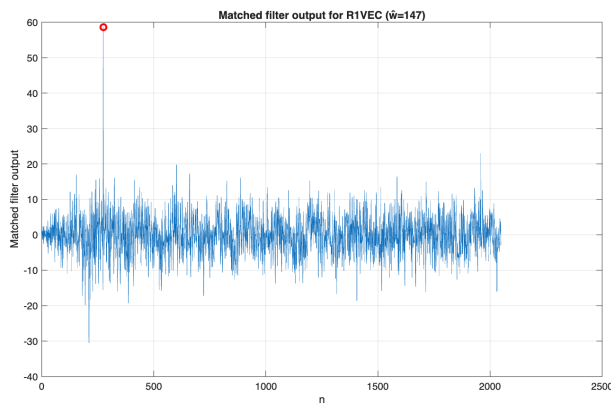
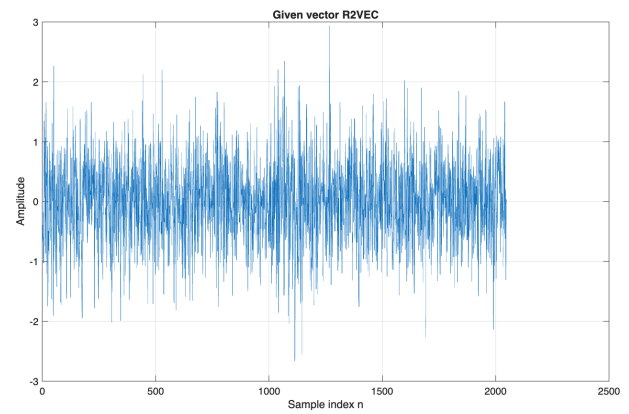
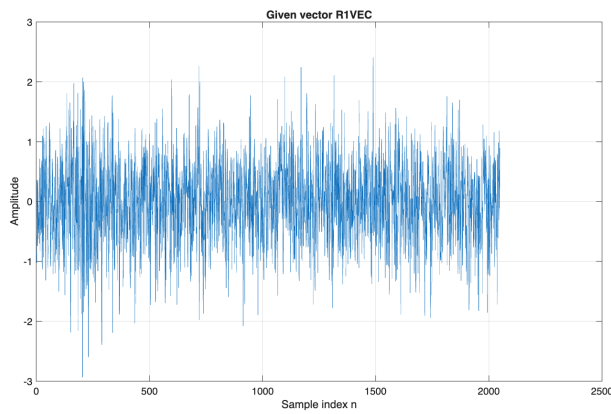
1. In order to estimate the unknown delay of the known signal a matched filter was applied. The matched filter was implemented by time-reversing the known signal and filtering the received vectors.

With $\delta = 1$ the signal appears so the MAP is the ML for the noise in this case.

The output of the matched filter shows a peak at the time index where the received signal best aligns with the known signal. Since MATLAB's filter function is causal, the peak index is shifted by the signal length. Therefore, the estimated delay was obtained by subtracting the signal length from the index of the maximum matched filter output.

Applying this method to the given vectors, resulted in the following delays:

- For $r_1[n]$: $\hat{w}_1 = 147$ samples
- For $r_2[n]$: $\hat{w}_2 = 1043$ samples



2. Main change in this question is the option the signal may not return.

Given vector $r \in R^M$ and known signal $s \in R^L$

We will define $H_0: r = n$ (just noise)

And define $H_w: r = s_w + n$, $w \in \{1, 2, \dots, M - L\}$ (signal started at w delay)

By the question $P(H_w) = \frac{\delta}{M-1}$, $P(H_0) = 1 - \delta$, $n \sim \mathcal{N}(0, \sigma^2 I)$

Using W_{map} as calculated in the lecture:

$$W_{MAP} = \begin{cases} \arg \max_{w \in [M-1]} t_w, & \text{if } t \geq \gamma \\ M & \text{otherwise} \end{cases}$$

$$\gamma = \frac{|s|^2}{2} + \sigma^2 \ln\left(\frac{1 - \delta}{\delta} (M - 1)\right)$$

Applying it to the question parameters we get.

$N = 2048$, $L = 128$, $M-1 = N-L = 1920 \rightarrow M = 1921$. $w \in \{1, 2, \dots, 1920\}$ and $w = 1921$ is no return for the signal.

$$\gamma = \frac{|s|^2}{2} + \sigma^2 \ln\left(\frac{1 - \delta}{\delta} (1920)\right)$$

$$W_{MAP} = \begin{cases} \arg \max_{w \in [1920]} t_w, & \text{if } t \geq \gamma \\ 1921 & \text{otherwise} \end{cases}$$

b. for r1:

$$\delta = 0.9, \sigma^2 = \frac{1}{2} \quad \text{Threshold} = 34.43 \quad \max MF = 48.38 \rightarrow \text{signal returned } \hat{w} = 147$$

For r2:

$$\delta = 0.9, \sigma^2 = 2.505$$

$$\text{Threshold} = 45.189$$

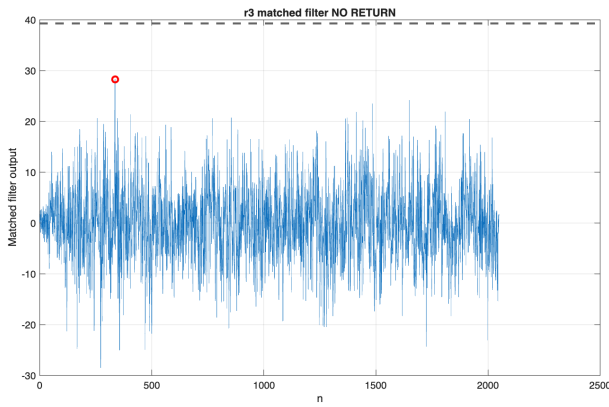
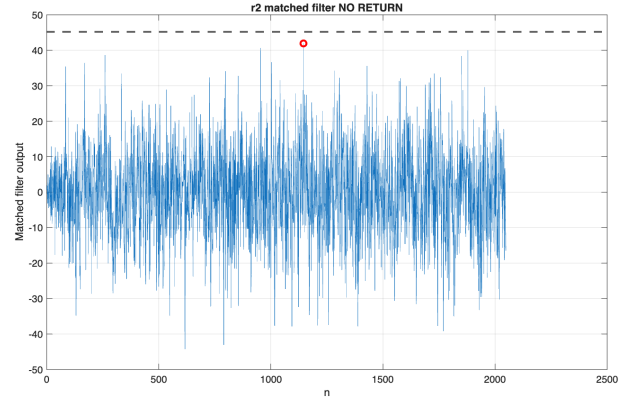
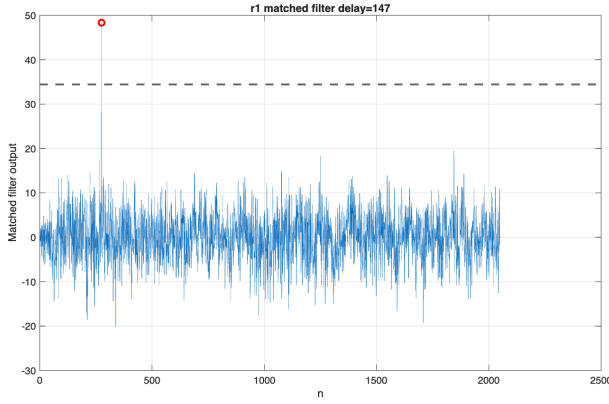
$$\max MF = 41.873 \rightarrow \text{signal didnt return because } \max MF < \text{Threshold}$$

For r3:

$$\delta = \frac{1}{2}, \sigma^2 = 0.9976$$

$$\text{Threshold} = 39.2922$$

$$\max MF = 28.2532 \rightarrow \text{signal didnt return because } \max MF < \text{Threshold}$$



3. $T_s = \frac{1}{F_s} = 2 * 10^{-7} [sec]$ $T = 4.096 * 10^{-4} [sec]$ $T * F_s = 2048 [\frac{samples}{pulse}]$

As in question number 1, we are here told that $\delta = 1$ which means the signal always returns. And that means ML = MAP.

$$Tower_1 = (0,0) \quad Tower_2 = (\Delta, 0) = (3840,0)$$

$$c = 3 * 10^8 [\frac{m}{sec}] \quad Delay = \tau = \frac{\omega}{F_s} \quad Path_1 = 2R_1 = c * \tau = c * \frac{\omega_1}{F_s} \rightarrow R_1 = c * \frac{\omega_1}{2F_s}$$

$$Path_2 = R_1 + R_2 = c * \frac{\omega_2}{F_s} \rightarrow R_2 = c * \frac{\omega_2}{F_s} - R_1$$

After knowing R_1 and R_2 we can calculate the spot of the airplane (x,y) by simple circle equation.

$$y = \sqrt{R_1^2 - x^2}, x = (R_1^2 - R_2^2 + \Delta^2)/2\Delta$$

Given the signal returning for each antenna I outputted a graph of the estimated location of the airplane by those simple calculations.

For each signal i: X_i and Y_i were calculated and the positions were dashed through the graph.

